
Active Causal Discovery Benchmark: Evaluating LLM Agents Under Budgeted Interventions

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We introduce the Active Causal Discovery Benchmark (ACDB), an SCM-grounded
2 environment for evaluating whether LLM agents recover causal graph structure
3 from observations and budget-constrained hard interventions. ACDB pairs a
4 linear-Gaussian world generator with a fixed observe-intervene-submit API and a
5 three-layer scoring contract that separates skeleton recovery, DAG recovery, and
6 intervention efficiency. On the current six-level ladder, PC with a greedy active
7 orientation heuristic is the strongest non-oracle method (directed F1 42.7%, SHD
8 4.79), ahead of Claude Sonnet 4.6 raw active (31.7%, 7.25) and GPT-5.4 raw active
9 (22.9%, 9.27). The most informative diagnostic is the precision-recall decom-
10 position: PC under-commits with high precision, LLMs over-commit with lower
11 precision, and statistical-tool access often increases abstention rather than useful
12 intervention. A structure-blind random DAG baseline reaches 23.6% directed F1
13 on this dense v0 ladder; a density probe lowers this floor to 16.9%, motivating the
14 v1 calibration pass. The current results should therefore be read as a benchmark
15 audit and calibration report, not as evidence that current LLMs solve active causal
16 discovery.

17 1 Introduction

18 Causal discovery — recovering the directed acyclic graph (DAG) behind an observed joint distribution
19 — is a prerequisite for many forms of causal reasoning. Classical methods such as PC and GES have
20 clean guarantees under faithfulness and causal sufficiency, but they are brittle at small samples and
21 do not by themselves specify how an agent should allocate a finite intervention budget.

22 Large language models (LLMs) are a natural stress case for this active setting. They can in prin-
23 ciple inspect data, choose tests, intervene, and decide when to submit; it remains unclear whether
24 this interaction produces reliable structure recovery or plausible-looking graphs driven by surface
25 regularities.

26 We ask:

27 *Can an LLM agent, given a budget of hard single-node interventions, recover a*
28 *causal DAG better than a structure-blind random baseline — and if so, by how*
29 *much, and under what conditions?*

30 To answer it we introduce **ACDB**, an Active Causal Discovery Benchmark. ACDB specifies a
31 linear-Gaussian SCM world generator, a push-based session API, a three-tier scoring contract, and a
32 six-level difficulty ladder. Every agent — classical or LLM — sees the same paired instances and is
33 scored by the same contract.

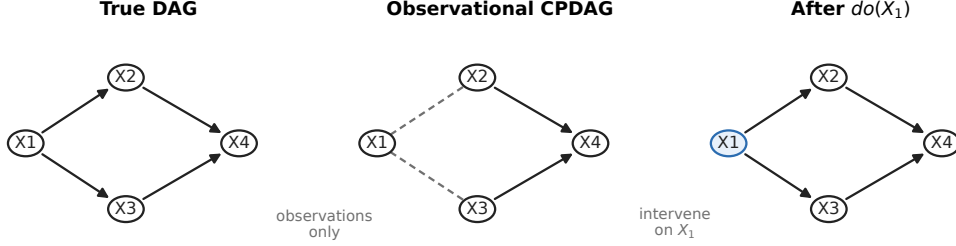


Figure 1: Observation identifies a CPDAG, not necessarily the full DAG. In this example the collider into X_4 is compelled from observational structure, while the two incident edges at X_1 remain undirected until an intervention on X_1 resolves them.

34 The current results support three findings. First, PC with a greedy active orientation heuristic is the
 35 strongest non-oracle method: directed F1 42.7%, SHD 4.79. Second, precision and recall expose
 36 different failure modes: PC under-commits with high precision (66.9%) and low recall (32.8%),
 37 whereas LLMs over-commit with lower precision and comparable recall. Third, a structure-blind
 38 random DAG baseline reaches 23.6% directed F1 on the dense current ladder, so weak LLM margins
 39 must be read against a high random floor.

40 Two secondary findings refine the picture. Statistical-primitive tools (correlation, partial correlation,
 41 Fisher- z tests) did not improve directed F1; they often shifted the model toward more testing
 42 and less intervention, producing high efficiency through abstention. At L2 and L5, Sonnet 4.6’s
 43 observational directed F1 exceeded its active-panel directed F1 on the same seeds, consistent with
 44 useful observational orientations being revised after intervention data.

45 The contribution is not a new causal discovery algorithm. It is a benchmark environment and a
 46 controlled empirical audit of how current LLM agents behave inside it. ACDB is intentionally narrow:
 47 causal sufficiency, linear-Gaussian SCMs, and perfect hard interventions. These restrictions make the
 48 scoring contract identifiable and give classical methods a fair comparison point. The density-leakage
 49 probe and full correlation-summary ladder define the next calibration work.

50 2 Background

51 We review only the formal machinery needed to define the benchmark and interpret its scores. Readers
 52 familiar with causal discovery may skip this section.

53 **Structural causal models.** A linear-Gaussian structural causal model (SCM) over $X_1, \dots, X_d$ is

$$X_i = \sum_{j \in \text{Pa}(i)} w_{ij} X_j + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2), \quad (1)$$

54 with $\text{Pa}(i)$ the parents of X_i in a DAG G and independent noise. The SCM induces an observational
 55 distribution $P(X)$ and, for every target (X_i, v) , an interventional distribution $P(X_{\neq i} \mid \text{do}(X_i = v))$
 56 obtained by replacing X_i ’s structural equation with $X_i := v$.

57 **Markov equivalence and the CPDAG ceiling.** Two DAGs are *Markov equivalent* iff they share
 58 the same skeleton and the same v-structures [Verma and Pearl, 1990, Andersson et al., 1997]. The
 59 equivalence class is represented by a completed partially directed acyclic graph (CPDAG), which
 60 orients all edges whose direction is identifiable from $P(X)$ (*compelled* edges) and leaves the rest
 61 undirected. An agent working from observations alone is bounded by the CPDAG; to recover the true
 62 DAG within an equivalence class, it must rely on structural assumptions or on interventions.

63 **Faithfulness and causal sufficiency.** *Faithfulness* states $X_i \perp\!\!\!\perp X_j \mid S \iff X_i$ is d -separated
 64 from X_j given S in G ; violations form a measure-zero subset of the parameter space but are
 65 empirically troublesome at finite samples, so our world generator rejects instances that approach the
 66 boundary. *Causal sufficiency* says that every common cause of observed variables is itself observed.
 67 v_0 assumes both, plus linear-Gaussian noise. These restrictions rule out latent confounders and

68 nonlinear mechanisms; they are deliberate because they make the scoring contract identifiable and
69 PC a meaningful baseline.

70 **The PC algorithm.** PC [Spirtes et al., 2000] tests conditional independences $X_i \perp\!\!\!\perp X_j \mid S$ for
71 increasing $|S|$, removes adjacencies, and orients edges via v-structure detection and Meek’s rules.
72 Under causal sufficiency, faithfulness, and a correct independence oracle, it returns the true CPDAG;
73 with finite samples it returns an empirical approximation dependent on the chosen test and significance
74 level.

75 An active causal discovery agent therefore has two distinct jobs: (i) recover the skeleton and
76 compelled orientations, a task bounded by $P(X)$; and (ii) resolve the remaining undirected edges
77 using interventions, a task that scales with the quality of the intervention policy. Our scoring contract
78 (§3) measures both separately.

79 3 The ACDB Benchmark

80 **ACDB** is an interactive benchmark environment in which an agent receives observational data, spends
81 a budget of hard single-node interventions, and submits an estimated causal graph. A benchmark
82 instance is $(G, \mathcal{M}, B, D^{\text{obs}})$: a ground-truth DAG, a linear–Gaussian SCM, an intervention budget,
83 and a fixed observational sample. The environment hides G and $\mathcal{M}$ and exposes only D^{obs} , d , and B .

84 3.1 World generation

85 Each instance assumes causal sufficiency, linear–Gaussian noise (Eq. 1), and edge weights drawn
86 uniformly from $[-2, -0.5] \cup [0.5, 2]$. We sample a uniform topological order, select exactly k forward
87 edges, and randomly permute node indices so names do not leak topology. Instances are rejected if
88 any relevant partial correlation falls below `faithfulness_eps` = 0.1. For each accepted instance
89 we precompute the minimum single-node intervention set $\mathcal{I}^*$ that orients every CPDAG-undirected
90 edge; this defines the efficiency baseline.

91 3.2 Session API

92 An ACDB session enforces a strict observe–intervene–submit protocol. `observe()` returns
93 the pre-sampled D^{obs} once. `intervene(var, value)` draws n_{int} fresh samples under the
94 hard intervention $\text{do}(X_{\text{var}} = \text{value})$ and decrements the budget; requests beyond B fail.
95 `submit_graph(submission)` seals the session. Submissions are mixed graphs of directed and
96 undirected edges; undirected edges represent orientations the agent was unable or unwilling to
97 commit to. Interventions are hard (the target variable is set exogenously; its parents are severed
98 during sampling), and the scoring contract treats an unresolved edge as a distinct failure mode from a
99 reversal or omission.

100 3.3 Scoring contract

101 ACDB reports three complementary views of a submission; all are deterministic given
102 $(G, \text{submission}, B)$.

103 **Observational layer: skeleton and compelled F1.** Against the true CPDAG of G , we compute
104 standard precision/recall/F1 on the undirected skeleton (*skeleton_f1*) and on the compelled (observa-
105 tionally orientable) edges (*compelled_f1*). A reversed compelled edge counts as both a false positive
106 on the reversed direction and a false negative on the true direction, so the agent is not rewarded for
107 “at least the adjacency was right” when it flipped a compelled orientation.

108 **DAG layer: directed F1 and SHD.** Against the true DAG G itself, we compute set-intersection
109 precision/recall/F1 on directed edges (*directed_f1*) and a structural Hamming distance *dag_shd*.
110 SHD counts, over the union of true and submitted adjacencies, the following errors at one per edge:
111 *extra* (submission has an edge, truth does not), *missing* (truth has an edge, submission has nothing),
112 *reversed* (both have a directed edge but in opposite directions), and *unresolved* (truth has a directed
113 edge, submission has an undirected edge on the same pair). Note that a reversal is one error, not two.

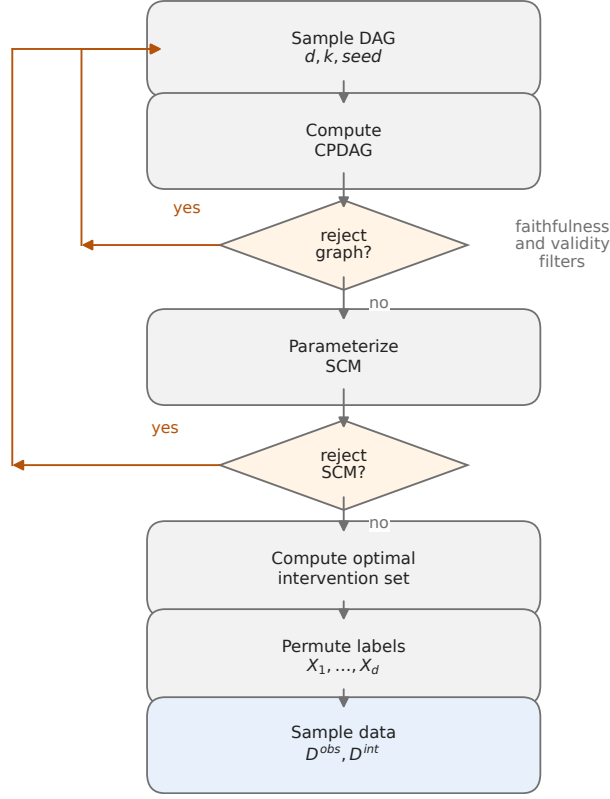
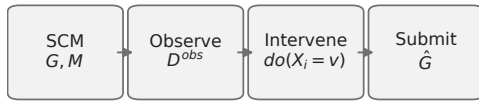


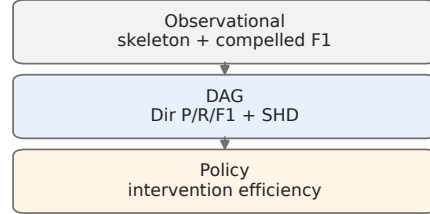
Figure 2: ACDB instance generation. Each accepted instance is produced by sampling a graph, converting it to its CPDAG, rejecting invalid or near-unfaithful cases, parameterizing a linear-Gaussian SCM, computing the optimal intervention set, anonymizing labels, and sampling data. Rejection gates loop back to graph sampling.

ACDB episode



same manifest, same API, same scoring

Scoring layers



Separates adjacency, orientation, and budget-use failures

Figure 3: ACDB fixes the episode protocol and separates adjacency, orientation, and budget-use scores.

114 **Efficiency layer.** Given the precomputed optimal intervention set $\mathcal{I}^*$ of size $|\mathcal{I}^*|$ and the number of
 115 interventions actually used, the efficiency score is

$$\eta = \frac{|\mathcal{I}^*|}{\max(|\mathcal{I}_{\text{used}}|, |\mathcal{I}^*|)}. \quad (2)$$

116 This is one whenever the agent used at most the optimal number of interventions (including zero-
 117 budget submission when $\mathcal{I}^* = \emptyset$), and decreases toward zero as wasted interventions accumulate.

Table 1: ACDB ladder. Each level fixes d (variables), k (true edge count), $n_{\text{obs}} / n_{\text{int}}$ (sample counts), σ^2 (noise variance), and `slack` (interventions above the precomputed optimal). The intervention budget is $B = |\mathcal{I}^*| + \text{slack}$ and is therefore instance-specific.

Level	Role	d	k	n_{obs}	n_{int}	σ^2	slack
0	tutorial	4	5	50	25	0.5	2
1	standard	5	6	25	15	1.0	1
2	statistical	5	6	15	10	1.5	1
3	structural	7	9	25	15	1.0	1
4	pressure	5	6	25	15	1.0	0
5	hard	7	9	15	10	1.5	0

118 **What the scoring contract excludes.** There is no MEC-aware partial credit in the DAG layer:
 119 the observational layer already expresses MEC-ceiling performance, and making the DAG layer
 120 MEC-aware would collapse the two.

121 3.4 The difficulty ladder

122 ACDB organizes instances into six levels, summarized in Table 1. Each level is a triple of difficulty
 123 axes: graph size and density, statistical power (sample counts and noise variance), and budget
 124 tightness (`budget_slack` controls how many interventions above $|\mathcal{I}^*|$ are allotted). The ladder is
 125 not a single monotone axis; levels 2 and 4 are designed to hold graph size fixed and isolate one axis
 126 at a time (statistical in L2, budget in L4), while L3 and L5 are structural stress levels.

127 Each level is evaluated on 8 seeds, preflight-validated so that instance assembly (DAG sample $\rightarrow$
 128 faithfulness check $\rightarrow$ SCM parameterization $\rightarrow$ optimal intervention set) succeeds for all seeds in
 129 the manifest. The same manifest is used across all agents and baselines for a given ladder run, so
 130 comparisons are paired.

131 3.5 Density leakage: a benchmark-design hazard

132 A structure-blind random baseline — one that samples a random DAG using only the public variable
 133 count d , with no access to the true edge count k — provides a floor for interpreting agent performance.
 134 When graph density $\rho = k / \binom{d}{2}$ is high, this floor is also high: a randomly sampled DAG agrees with
 135 the truth on a non-trivial fraction of directed edges simply because both contain many edges. We call
 136 this effect *density leakage*.

137 Empirically, on the ladder of Table 1, the structure-blind random baseline attains overall directed F1
 138 of 0.236 across 4800 samples (Section 5). Measuring the same baseline at several post hoc densities,
 139 without giving the baseline access to k , we observe an approximately monotone heuristic:

$$\mathbb{E}[\text{Dir-F1}_{\text{random}}] \approx \frac{\rho}{1 + 2\rho}, \quad \rho = \frac{k}{\binom{d}{2}}. \quad (3)$$

140 Equation (3) is a probe fit, not a derived bound. The density-probe ladder ($\rho \leq 0.33$ on non-tutorial
 141 levels) reduces the measured random floor from 23.6% to 16.9% overall. We keep the original ladder
 142 for v0 because the LLM and PC runs were collected there, and use the probe as a concrete target for
 143 v1 calibration.

144 4 Agents and Baselines

145 All agents interact through the same session API (§3), see the same manifest of 8 preflighted seeds
 146 per level, and are scored by the same three-tier contract. Agents differ only in the policy inside the
 147 session.

148 **PC and PC + greedy active.** The observational baseline runs PC from the `causal-learn` library
 149 on D^{obs} with Fisher- z independence tests at $\alpha = 0.05$ and submits the resulting CPDAG. The
 150 active baseline extends this procedure: after PC, it repeatedly selects the node with the highest

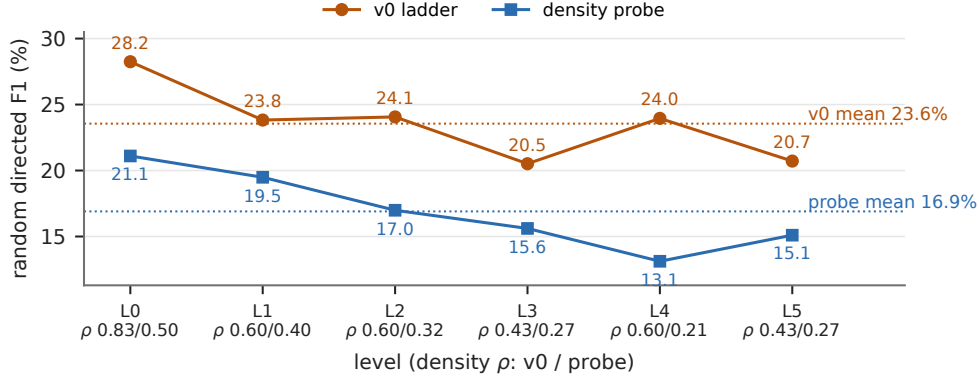


Figure 4: Density leakage in the structure-blind random baseline. Points are per-level directed F1; tick labels show graph density ρ for the v0 ladder and density probe. Lowering density reduces the overall random floor from 23.6% to 16.9%.

undirected-edge degree, intervenes at $\text{do}(X_i = \bar{X}_i + 3.0)$ where $\bar{X}_i$ is the observational mean, and orients each unresolved neighbor by comparing the post-intervention mean against the observational mean (shift threshold 0.5 in absolute value). Cycles are rejected. The greedy heuristic is simple and not optimal, but it provides a clear reference policy for the decisions an LLM policy must make. An *oracle* baseline submits the true DAG as an upper bound.

Structure-blind random baseline. The random-uniform baseline quantifies how much directed-F1 signal is achievable without using data. Using only the public variable count d (not D^{obs} , not k , no interventions), it samples a random topological order, enumerates the $M = d(d-1)/2$ forward edges, draws an edge count m uniformly from $\{0, \dots, M\}$, and submits m random forward edges. We run 100 such samples per instance. Results near this floor should not be interpreted as evidence of reliable causal discovery (§5.3).

LLM agents. Two backbones (GPT-5.4 and Claude Sonnet 4.6), four configurations each: raw (`llm_raw`) vs stats (`llm_stats`), on both observational and active panels. Temperature is 0. Each turn the model receives the task prompt, a static session prompt (variable names, observational matrix, initial budget), and the full tool-call history, and emits exactly one tool call from `{intervene, correlation, partial_correlation, independence_test, submit_graph}`. Parallel tool calls are disabled. Episodes end at `submit_graph` or at the step limit (16 for raw, 32 for stats). The raw variant must reason over the raw data matrix; the stats variant additionally has correlation, partial-correlation, and Fisher- z tools. The raw/stats comparison tests whether handing the model statistical primitives closes the gap to PC; the observational/active split measures whether the model can use interventions beyond the CPDAG ceiling.

Correlation-only ablation probe. A restricted-prompt probe (`llm_corr_obs`) gives the model only the correlation matrix (rounded to three decimal places) and sample standard deviations, with no raw data, no tools, no interventions, and a system prompt that warns that correlation does not imply causation and asks the model to withhold orientation when uncertain. The model must respond with a single `submit_graph` call. We run this probe on levels $\{1, 3, 5\}$ with 4 seeds per level and both backbones. It is a diagnostic ablation, not a primary comparison (§5.4).

5 Results

We evaluate three families of agents on the ladder of Table 1: the PC algorithm (classical constraint-based baseline), LLM policies (GPT-5.4 and Claude Sonnet 4.6, each in raw-data and statistical-tools variants), and two reference probes (a structure-blind random DAG baseline and a correlation-only LLM ablation). The run comprised 336 planned jobs per model across the full ladder; GPT-5.4 completed all 336 successfully, Sonnet 4.6 completed 335 (one `llm_stats` active row failed). All

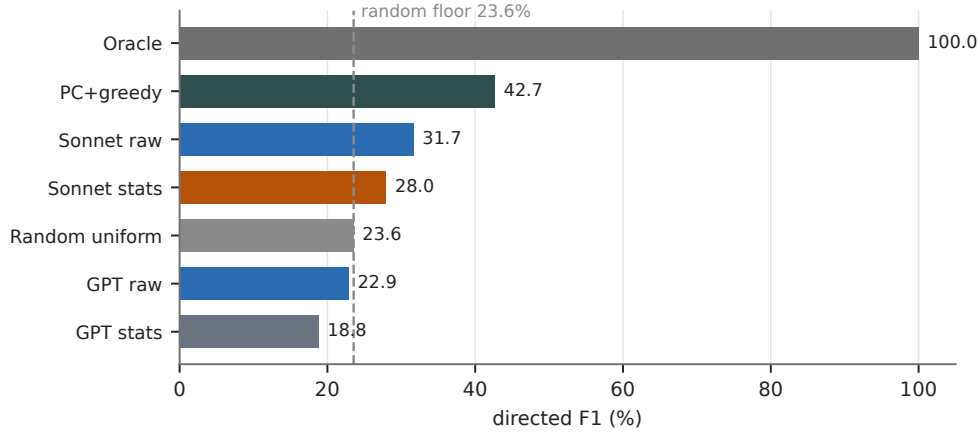


Figure 5: Headline DAG-recovery performance on the v0 active ladder. PC+greedy is the strongest non-oracle method, Sonnet raw is the strongest LLM policy, and several LLM settings sit close to the structure-blind random floor.

ladder-evaluated agents share the same manifest of 8 seeds per level, so every pairwise comparison is on identical instances.

5.1 Headline results

Table 2 reports weighted averages across L0–L5 with directed precision and recall broken out alongside F1. Paired sign-flip tests on deduplicated same-seed rows support the main active comparisons: PC+greedy beats the best LLM on directed F1 by 11.0 points ($p = 0.025$) and SHD by 2.46 edges ($p < 10^{-4}$), while Sonnet raw beats GPT raw in the active panel by 8.8 points ($p = 0.028$). Four findings summarize the run.

Finding 1: PC with active orientation is strongest on the primary metrics. PC + greedy achieved directed F1 of 42.7% and SHD of 4.79, outperforming every LLM agent. The best LLM (Claude Sonnet 4.6 active raw) reached directed F1 of 31.7% and SHD of 7.25 under the same budget. The SHD gap is the more robust of the two because it is less sensitive to aggressive guessing.

Finding 2: the P/R decomposition reveals qualitatively different errors. On directed edges in the active panel, PC + greedy had precision 66.9% and recall 32.8%: it commits to few edges but is usually correct when it does. Sonnet 4.6 raw was more balanced (35.0/29.6), while GPT-5.4 raw submitted aggressively with low precision (36.3/19.3). This is the main qualitative finding: *PC under-commits, LLMs over-commit, and F1 alone collapses both errors into a single number.* Reporting only F1 would understate how different the two failure modes are.

Finding 3: Sonnet 4.6 is stronger than GPT-5.4 in every overall LLM setting. Directed F1 improved for Sonnet 4.6 across raw and stats, observational and active. The gap was largest on raw observational (36.1% vs 22.6%, $p = 0.0028$) and raw active (31.7% vs 22.9%, $p = 0.028$). Under these prompts and seeds, Sonnet 4.6’s observational panel matched or exceeded every GPT-5.4 setting, including active, on directed F1.

Finding 4: statistical-tool access shifted behavior toward abstention. Agents given correlation, partial correlation, and Fisher- z tools did not outperform raw-data agents on directed F1: Sonnet raw beat Sonnet stats (31.7% vs 28.0%), and GPT raw beat GPT stats (22.9% vs 18.8%). Stats agents ran longer trajectories, intervened less, and submitted fewer directed edges; their high efficiency therefore reflects abstention rather than better active discovery.

5.2 Per-level breakdown

Table 3 shows directed F1 per level for both panels. Three patterns are visible.

Table 2: Headline results as weighted averages across ladder levels L0–L5. Skel F1 is against the true CPDAG; Dir P, Dir R, Dir F1 are against the true DAG; SHD is edge edit distance (reversal=1); η is intervention efficiency against the precomputed optimal set. All values in %; SHD is a mean edge count. Oracle submits the true DAG; PC + greedy is the strongest non-oracle active method. Directed precision and recall are reported separately because LLM and PC errors lie on different sides of the P/R trade-off (§5).

Method	Model	Skel F1	Dir P	Dir R	Dir F1	SHD	η
<i>Observational panel (no interventions)</i>							
PC	–	62.1	82.6	10.0	13.6	6.25	–
LLM (raw)	GPT-5.4	65.7	74.1	19.2	22.6	9.40	–
LLM (stats)	GPT-5.4	35.9	68.4	9.2	13.3	6.75	–
LLM (raw)	Sonnet 4.6	57.7	38.9	34.5	36.1	7.46	–
LLM (stats)	Sonnet 4.6	60.7	76.9	12.5	15.0	9.92	–
<i>Active panel (hard-intervention budget)</i>							
Oracle	–	100.0	100.0	100.0	100.0	0.00	100.0
PC + greedy	–	62.1	66.9	32.8	42.7	4.79	89.6
LLM (raw)	GPT-5.4	58.8	36.3	19.3	22.9	9.27	92.7
LLM (stats)	GPT-5.4	40.0	48.0	13.3	18.8	6.58	100.0
LLM (raw)	Sonnet 4.6	56.3	35.0	29.6	31.7	7.25	97.9
LLM (stats)	Sonnet 4.6	62.7	46.0	22.7	28.0	8.06	100.0
<i>Reference points</i>							
Random uniform DAG	–	46.6	35.0	25.0	23.6	8.18	–
Correlation-only probe [†]	GPT-5.4, Sonnet 4.6	~70	–	–	0.0	–	–

[†]Prompt-restricted ablation: single-call graph submission from the correlation matrix only, with no interventions or tool calls. Skeleton F1 $\approx$ 70%; directed F1 = 0 under the prompt restriction, because the models withheld orientation when uncertain. Not a primary comparison. See Section 5.4.

Table 3: Directed F1 per ladder level, by panel. Weighted averages in the rightmost column match Table 2. Values are means over 8 preflighted seeds. The random row is structure-blind and uses only d . All values in %.

Method	Model	L0	L1	L2	L3	L4	L5	Overall
<i>Observational panel</i>								
PC	–	13.9	23.8	3.1	17.2	20.0	3.8	13.6
LLM (raw)	GPT-5.4	39.8	27.7	7.5	18.8	24.4	17.7	22.6
LLM (stats)	GPT-5.4	12.3	13.4	9.4	11.4	21.9	11.8	13.3
LLM (raw)	Sonnet 4.6	26.9	34.9	39.3	33.8	41.7	40.1	36.1
LLM (stats)	Sonnet 4.6	24.7	14.0	6.2	10.9	22.4	12.0	15.0
<i>Active panel</i>								
Oracle	–	100.0	100.0	100.0	100.0	100.0	100.0	100.0
PC + greedy	–	65.1	49.7	30.7	48.9	46.5	15.2	42.7
LLM (raw)	GPT-5.4	28.5	24.1	27.6	20.6	15.0	21.6	22.9
LLM (stats)	GPT-5.4	31.2	13.4	12.2	18.3	21.6	16.4	18.8
LLM (raw)	Sonnet 4.6	29.4	28.4	36.9	29.7	33.6	31.9	31.7
LLM (stats)	Sonnet 4.6	26.4	33.2	29.7	28.7	26.5	23.5	28.0
Random uniform	–	28.2	23.8	24.1	20.5	24.0	20.7	23.6

214 First, PC + greedy is strongest overall but not uniformly across levels. Sonnet raw exceeds it on L2
215 (36.9% vs 30.7%) and L5 (31.9% vs 15.2%), the two low-sample levels where PC’s finite-sample
216 errors are largest.

217 Second, Sonnet 4.6 is the stronger LLM in the overall aggregates, but not every individual cell.
218 GPT-5.4 occasionally lands near the random floor, especially in the observational panel at L2 and L5.

219 Third, the *observational* panel outscores the *active* panel for Sonnet 4.6 raw at L2 (39.3% vs 36.9%)
220 and L5 (40.1% vs 31.9%). These paired gaps are not significant at eight seeds ($p = 0.50$, $p = 0.20$),
221 but they are diagnostically useful: intervention data did not reliably improve the policy on the hardest
222 levels.

223 5.3 Density leakage: the random floor

224 The structure-blind random baseline attained overall directed F1 of 23.6% on the ladder, with per-
225 level values from 28.2% at L0 down to about 21% at structural levels. Several GPT-5.4 settings land
226 near this floor (raw active 22.9%, raw observational 22.6%, stats active 18.8%), so their directed-F1
227 margins should be treated cautiously. A separate density probe, using the same random policy
228 without giving it k , reduces the random floor to 16.9% overall. This is the empirical reason v1 should
229 recalibrate density before drawing stronger conclusions about LLM capability.

230 5.4 Correlation-only ablation probe

231 As a diagnostic, we ran `llm_corr_obs` on L1/L3/L5 with 4 seeds per level and both backbones.
232 The model saw only standard deviations and a rounded correlation matrix, no raw data, tools,
233 or interventions. Skeleton F1 was non-trivial (67–74% by level/model), showing that compact
234 summaries carry adjacency signal in this SCM family. Directed F1 was 0 because the prompt
235 explicitly asked the models to withhold unsupported orientations and they submitted undirected edges.
236 This is an incomplete v0 ablation, not a primary DAG-recovery comparison.

237 5.5 Efficiency: budgets were rarely the binding constraint

238 PC + greedy achieved efficiency $\eta = 89.6\%$; it sometimes used redundant interventions. LLM
239 agents achieved $\eta \in [92.7, 100.0]\%$, with both `llm_stats` settings at exactly 100.0%. This is not
240 skilled budget management: the stats agents rarely intervened, preferring statistical tool calls and
241 conservative submissions. High efficiency without matching precision/recall gains is evidence of
242 abstention.

243 5.6 Summary

244 PC + greedy is strongest overall because it combines high directed precision with a narrow commit
245 pattern. LLMs commit more aggressively and with lower precision. Sonnet 4.6 is the stronger overall
246 LLM under these prompts, statistical-tool access does not improve directed F1, and current LLM
247 policies do not reliably convert intervention budget into orientation accuracy.

248 6 Discussion

249 We interpret the main findings and identify the calibration work needed for the next benchmark
250 version.

251 6.1 What the results say about current LLMs

252 **Classical methods still win on clean formal tasks.** PC + greedy achieved the highest directed F1
253 and the lowest SHD on the ladder. Under the v0 assumptions, this is consistent with theory: PC is
254 asymptotically correct and the greedy orientation heuristic is strong enough for small graphs. Current
255 LLM agents are not drop-in replacements for classical structure-learning algorithms in settings where
256 the SCM lies in their identifiability regime.

257 **The precision–recall asymmetry is the main behavioral signal.** PC under-commits (precision
258 66.9, recall 32.8); LLMs over-commit (precision 35–46, recall 20–30). Neither failure mode is
259 uniformly preferable: under-committers leave useful edges unresolved, while over-committers add
260 wrong orientations. A benchmark that reports only F1 collapses this distinction. ACDB’s scoring
261 contract keeps precision and recall visible.

262 **Interventions sometimes hurt.** At L2 and L5, Sonnet 4.6’s observational directed F1 exceeded
263 its active directed F1 on identical seeds (39.3% vs 36.9%; 40.1% vs 31.9%). Interventions are more
264 informative than observations, so this is a policy diagnostic: current prompts do not reliably convert
265 intervention samples into better orientation decisions. Edge-level revision analysis is a v1 target.

266 **Statistical tools did not improve active discovery.** Giving LLMs correlation, partial-correlation,
267 and independence-test tools did not raise directed F1 above raw data. The stats variants ran longer
268 trajectories, intervened less, and submitted fewer directed edges; their 100% efficiency reflects
269 abstention, not skill. Tool access alone is insufficient if the model does not know when to stop testing
270 and commit.

271 6.2 What the results say about the benchmark

272 **Density leakage is an open calibration issue.** The random baseline’s 23.6% directed F1 on the
273 current ladder is high enough that several GPT-5.4 settings sit near it. The density probe reduces the
274 random floor to 16.9% overall, turning the ladder issue into a concrete v1 calibration target rather
275 than a vague caveat. The v0 results remain useful because all agents share paired seeds on the same
276 ladder.

277 **The scoring contract separates distinct failure modes.** The decomposition into skeleton,
278 compelled-edge, DAG, SHD, and efficiency scores separated the CPDAG-ceiling question from
279 the orientation-under-intervention question from the policy-quality question. The correlation-only
280 probe would not be legible in an F1-only benchmark: its zero directed F1 looks like a failure, but
281 the prompt-restricted reading — the model withheld orientation when uncertain — is only available
282 because skeleton F1 is reported alongside.

283 6.3 v1 and future variants

284 The next benchmark version should keep the same SCM foundation but fix the density ladder, run the
285 full LLM-correlation summary condition, increase seeds, and inspect intervention-driven revision
286 behavior. Beyond that, the same generator/API can support causal inference with a known DAG,
287 counterfactual queries, decision-making under causal uncertainty, transfer across SCM families, and
288 RL for experimental design. Those are extensions of the scaffold, not claims made by this v0 run.

289 6.4 Limitations

290 Four limitations are central to interpreting the results. (1) v0 assumes causal sufficiency and linear-
291 Gaussian SCMs. (2) The ladder has density-leakage artifacts at the dense end. (3) Eight seeds per
292 level is adequate for coarse trends but under-powered for fine-grained ranking. (4) We tested two
293 LLM backbones and one prompt design each, so LLM capability claims are conditional on those
294 choices.

295 7 Related Work

296 **Classical causal discovery.** ACDB builds on the PC algorithm [Spirtes et al., 2000], Markov-
297 equivalence theory [Verma and Pearl, 1990, Andersson et al., 1997], score-based GES [Chickering,
298 2002], and intervention-design work on orienting equivalence classes [Eberhardt et al., 2005, Hauser
299 and Bühlmann, 2012]. These methods define the CPDAG ceiling and the classical baselines against
300 which LLM agents should be evaluated.

301 **LLMs for causality.** Prior LLM work ranges from optimistic pairwise causal-direction results
302 [Kiciman et al., 2023] to critiques that such behavior is pattern matching rather than causal reasoning
303 [Zečević et al., 2023]. CLadder evaluates language-model causal QA across Pearl’s ladder [Jin et al.,
304 2023], while Long et al. [2023] test text-only graph reconstruction. ACDB differs by requiring the
305 model to interact with data and interventions before committing to a scored graph.

306 **Agent benchmarks.** SWE-bench, AgentBench, WebArena, OSWorld, and GAIA move evaluation
307 toward interactive agents [Jimenez et al., 2024, Liu et al., 2024, Zhou et al., 2024, Xie et al., 2024,
308 Mialon et al., 2023]. ACDB shares the paired-seed, deterministic-scoring ethos but narrows the task
309 to scientific reasoning under a formal SCM, where the hidden state and failure modes are inspectable
310 by construction.

8 Conclusion

We asked whether an LLM agent, given budgeted hard interventions, can recover a causal DAG better than a structure-blind random baseline. On the v0 ladder, PC + greedy is strongest overall (42.7% directed F1, SHD 4.79); the best LLM reaches 31.7% directed F1 with SHD 7.25; and the random baseline reaches 23.6%, high enough that weak LLM margins require caution. The precision–recall decomposition shows qualitatively different errors: PC under-commits with high precision, LLMs over-commit with lower precision, and stats-tool variants often abstain from intervention. ACDB is therefore a scaffold for tracking active causal discovery as models and benchmark calibration improve. The next version should lower the random floor, run the full correlation-summary condition, and inspect intervention-driven revisions.

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